The argument is moving towards the idea of a stereotyped, regularly repeating life cycle. Not only does each generation begin with a single-celled bottleneck. It also has a growth phase—'childhood'—of rather fixed duration. The fixed duration, the stereotypy, of the growth phase makes it possible for particular things to happen at particular times during embryonic development, as if governed by a strictly observed calendar. To varying extents in different kinds of creature, cell divisions during development occur in rigid sequence, a sequence that recurs in each repetition of the life cycle. Each cell has its own location and time of appearance in the roster of cell divisions. In some cases, incidentally, this is so precise that embryologists can give a name to each cell, and a given cell in one individual organism can be said to have an exact counterpart in another organism.

So, the stereotyped growth cycle provides a clock, or calendar, by means of which embryological events may be triggered. Think of how readily we ourselves use the cycles of the earth's daily rotation, and its yearly circumnavigation of the sun, to structure and order our lives. In the same way, the endlessly repeated growth rhythms imposed by a bottlenecked life cycle will—it seems almost inevitable—be used to order and structure embryology. Particular genes can be switched on and off at particular times because the bottleneck/growth-cycle calendar ensures that there *is* such a thing as a particular time. Such well-tempered regulations of gene activity are a prerequisite for the evolution of embryologies capable of crafting complex tissues and organs. The precision and complexity of an eagle's eye or a swallow's wing couldn't emerge without clockwork rules for what is laid down when.

The third consequence of a bottlenecked life history is a genetic one. Here, the example of bottle-wrack and splurge-weed serves us again. Assuming, again for simplicity, that both species reproduce asexually, think about how they might evolve. Evolution requires genetic change, mutation. Mutation can happen during any cell division. In splurge-weed, cell lineages are broad-fronted, the opposite of bottlenecked. Each branch that breaks apart and drifts away is many-celled. It is therefore quite possible that two cells in a daughter will be more distant relatives of one another than either is to cells in the parent plant. (By 'relatives', I literally mean cousins, grandchildren and so on. Cells have definite lines of descent and these lines are branching, so words like second cousin can be used of cells in a body without apology.) Bottle-wrack differs sharply